

CSE 4304-Data Structures Lab. Winter 2024-25

Date: 25 February 2026

Target Group: All groups

Topic: Range Queries: Prefix Sum, Sparse Table, and Fenwick Tree.

Instructions:

- Task naming format: fullID_T01L01_2A.c/cpp
- If you find any issues in the problem description/test cases, comment in the Google Classroom.
- If you find any tricky test cases that I didn't include but that others might forget to handle, please comment! I'll be happy to add them.
- Use appropriate comments in your code. This will help you recall the solution more easily in the future.
- The obtained marks will vary based on the efficiency of the solution.
- Do not use `<bits/stdc++.h>` library.
- Modified sections will be marked with **BLUE** color.
- You are allowed to use the STL stack unless it's specifically mentioned to use manual functions.

Group	Tasks
1A/1B/2A/2B	1 2 3
Assignment	

Task 1: Find the Most Common Character in a Substring Range

Given a string S of length N and Q queries, each consisting of two integers L and R (1-indexed), representing a substring $S[L, R]$.

Input:

The first line provides the string. The second line carries the value of Q . The following Q lines provide the values of L and R representing the ranges.

For each query, output the most common character in the substring. If there is a tie, return the lexicographically smallest character.

Sample Input	Sample output	Explanation
abcccabaaabb 3 1 4 2 5 6 12	c:2 c:3 a:4	

Note: Use prefix sum to compute this.

Task 2: Implement the basic operations of a Sparse Table

Given an array of N integers, find the minimum value of a range [L, R] using the Sparse Table data structure.

You have to print the following information:

1. The ranges to be stored in the Sparse table
2. Print the Sparse table after construction
3. Answer the result of each query and how the answer was extracted from the Sparse Table

Input:

The first line contains the value of N. The Second line provides the N integers.

The third line will contain the Q value, representing the number of queries. The following Q lines will provide the range definitions.

Sample Input	Sample output
8 2 3 4 5 3 1 1 3 5 2 6 1 3 1 7 2 7 3 4	Ranges to be stored in Sparse Table: (0,0) (0,1) (0,3) (0,7) (1,1) (1,2) (1,4) (2,2) (2,3) (2,5) (3,3) (3,4) (3,6) (4,4) (4,5) (4,7) (5,5) (5,6) (6,6) (6,7) (7,7) Status of Sparse Table: 2 2 2 1 3 3 3 4 4 1 5 3 1 3 1 1 1 1 1 1 3 Query-1: Min=1 min([2,5],[3,6]) Query-2: Min=3 min([1,2],[2,3]) Query-3: Min=1 min([1,4],[4,7]) Query-4: Min=1 min([2,5],[4,7]) Query-5: Min=3 min([3,4],[3,4])

Task 3: Implement the basic operations of a Fenwick Tree

Given an array of N integers, find the sum of a range [L, R] using the Fenwick Tree (a.k.a Binary Indexed Trees) data structure.

You have to print the following information:

- The ranges to be stored in the Fenwick Tree
- Print the Tree after construction (used linear construction)
- Answer the result of each query and how the answer was extracted from the Tree.

Input:

- The first line contains the value of N. The Second line provides the N integers.
- The third line will contain the Q value, representing the number of queries. The following Q lines will provide the range definitions.

Sample Input	Sample output
16 5 10 5 10 5 10 5 10 5 10 5 10 5 10 5 10 3 1 7 1 11 11 15	Ranges to be stored in the Fenwick Table: 1:(1,1) 2:(1,2) 3:(3,3) 4:(1,4) 5:(5,5) 6:(5,6) 7:(7,7) 8:(1,8) 9:(9,9) 10:(9,10) 11:(11,11) 12:(9,12) 13:(13,13) 14:(13,14) 15:(15,15) 16:(1,16) Status of Fenwick Tree (idx: value): 1:5 2:15 3:5 4:30 5:5 6:15 7:5 8:60 9:5 10:15 11:5 12:30 13:5 14:15 15:5 16:120 Query-1: Sum=50 [Tree idx: sum(7,6,4) - sum(1)] Query-2: Sum=80 [Tree idx: sum(11,10,8) - sum(1)] Query-3: Sum=35 [Tree idx: sum(15,14,12,8) - sum(11,10,8)]

Note: We'll cover the point update in the assignment.